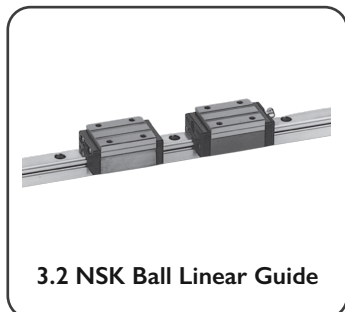


NSK Product Brochure

V2021JAN



3.1 NSK Roller Linear Guide



3.2 NSK Ball Linear Guide



3.3 NSK Ball Screw



3.4 NSK Roller Pack



3.5 NSK Special Grease



3.6 NSK Hand Grease Pump



3.8 NSK SPACEA™ Series



3.9 NSK Super Precision Bearings



3.11 NSK Roller Bearings



3.7 NSK Nozzels



3.10 NSK Grease

NSK Ltd. (日本精工株式会社 **Nippon Seikō Kabushiki-geisha, Japanese Precision Company**) is a manufacturer of Bearings, Linear Guide, Ball Screw and lots of precision products globally and the largest in Japan. It is a 104 years old Japanese Multinational Company. They are world leaders, with more than 27% market share, for the CNC and machine tools applications Ball Screws. The company produces industrial machinery Bearings like super precision Bearings and SPACEA Bearings, precision machinery and parts like Linear Guides and Ball Screws, and automotive Bearings and components.

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3.1 Roller Linear Guide

NSK Roller Linear Guide has two basic types, Which are Square and Flange type. Roller Guides are preferred for long life and impact loading conditions.



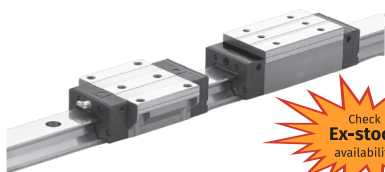
Industries it caters to

Automobile Industry
Polyfilms Industry
Packaging Industry

Product application

JSW Injection Molding Machine
UBE Injection Molding
Toshiba
Duplex Milling Cutter

3.1.1 SQUARE TYPE (AN,BN,AL,BL)



Check
Ex-stock
availability

Selection parameters

Parameters	Min	Max
Rail Width	15mm	65mm
Assembly Height	28mm	90mm
Dynamic Load	10300 N	288000 N
Static Load	27500 N	756000 N

3.1.2 FLANGE TYPE (EM,GM)



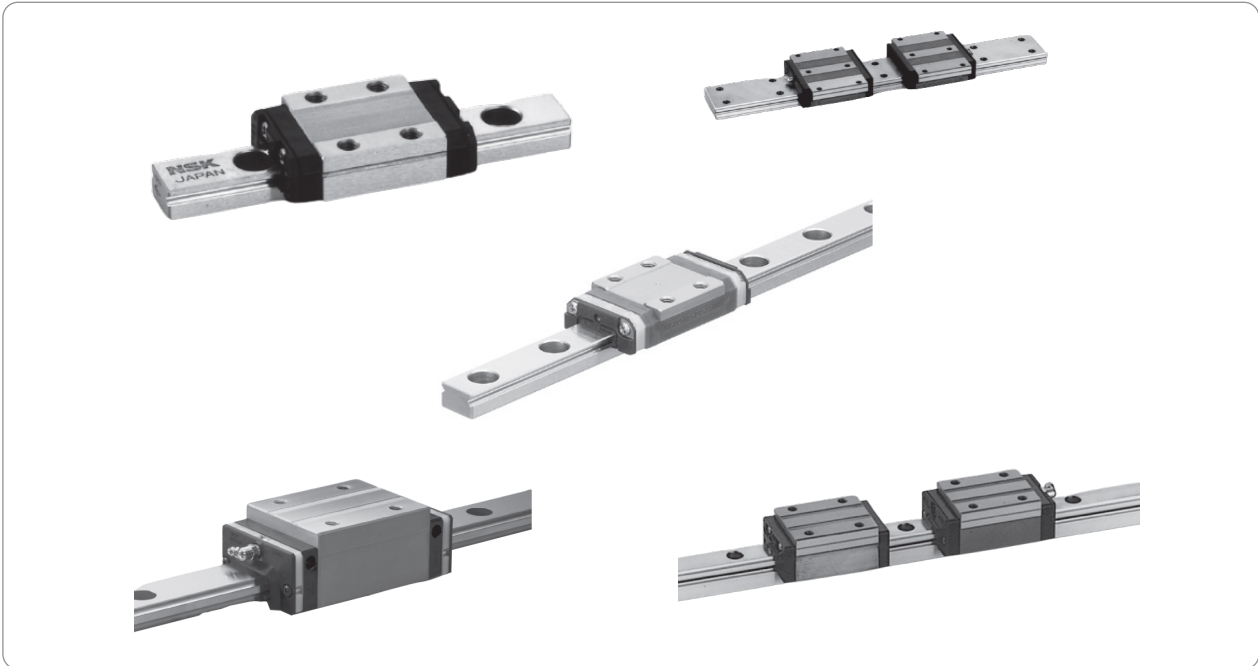
Check
Ex-stock
availability

Selection parameters

Parameters	Min	Max
Rail Width	15mm	65mm
Assembly Height	24mm	90mm
Dynamic Load	10300 N	288000 N
Static Load	27500 N	756000 N

3.2 Ball Linear Guide

NSK has designed new ball groove geometry for Ball Linear Guide. It features improved rated life and rolling direction. Accuracy and impact load resistance is highly improved. There are many variants like NH series, NS series, etc. used in different applications like Laser Cutting Machines, Material Handling Equipment.



Industries it caters to

Packaging Equipment Manufacturer
Material Handling Equipment Manufacturer
Electronic Part Manufacturer
Industrial Robots
Woodworking Machines
Laser Cutting Machines
Electric Discharge Machines
Medical Equipment Manufacturer
Measuring Equipment Manufacturer
Printing Equipment's Manufacturer
LCD Manufacturing Equipment
Paper Manufacturing
Packaging/Packing Machines
Miniature Robots
PCB-Drilling Machine

Product application

NTC HMC Machine N4Q Model
Lasser Embroidary Machine
Widos Butt Fusion Welding Machine
Brothers- Japan CNC Machine
Jyoti CNC Machine
Others

3.2 Ball Linear Guide (continued)

3.2.1 NH SERIES



NSK has designed new ball groove geometry for Ball linear guide. It features improved rated life and rolling direction. Accuracy and impact load resistance are highly improved. There are many variants like NH series, NS series etc. used in different applications like Laser cutting machines, material handling equipments.

Selection parameters

Parameters	Min	Max
Rail Width	15mm	65mm
Assembly Height	24mm	90mm
Dynamic Load	10300 N	288000 N
Static Load	27500 N	756000 N

3.2.2 LW SERIES



NSK has developed LW series for moment loading. This is being a wide rail suits for single rail systems, hence makes it compact for application.

Selection parameters

Parameters	Min	Max
Rail Width	33mm	90mm
Assembly Height	17mm	50mm
Dynamic Load	4450 N	48500 N
Static Load	11300 N	117000 N

3.2.3 PU SERIES



NSK has designed miniature Linear Guides like PU series. It is lightweight and operates smoothly. Dust generation and noise are reduced. It also has good corrosion resistance.

Selection parameters

Parameters	Min	Max
Rail Width	5mm	15mm
Assembly Height	6mm	16mm
Dynamic Load	440 N	6400 N
Static Load	775 N	11300 N

3.2.4 LU SERIES



NSK LU series is a super small linear guide. It has gothic arc groove on both sides. It can take equal loads in vertical and horizontal directions.

Selection parameters

Parameters	Min	Max
Rail Width	5mm	15mm
Assembly Height	6mm	16mm
Dynamic Load	435 N	6400 N
Static Load	740 N	11300 N

3.2.5 PE SERIES



NSK PE Series is a miniature wide type Linear Guide. The features like lightweight, dust proofing, low noise makes it easy to operate and compact.

Selection parameters

Parameters	Min	Max
Rail Width	10mm	42mm
Assembly Height	6.5mm	16mm
Dynamic Load	550 N	8200 N
Static Load	1160 N	16000 N

3.2.6 LE SERIES



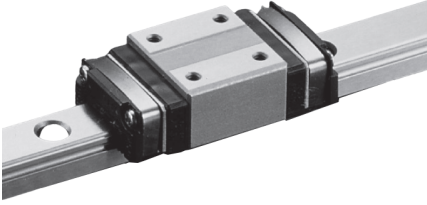
NSK LE is miniature Linear Guide with wide rail. These are super thin Linear Guides. It is ideal for single rail use.

Selection parameters

Parameters	Min	Max
Rail Width	10mm	42mm
Assembly Height	6.5mm	16mm
Dynamic Load	470 N	8200 N
Static Load	835 N	16000 N

3.2 Ball Linear Guide (continued)

3.2.7 MINIATURE LH SERIES



NSK Miniature LH Series features very good self aligning capacity, high impact load and corrosion resistance.

Selection parameters

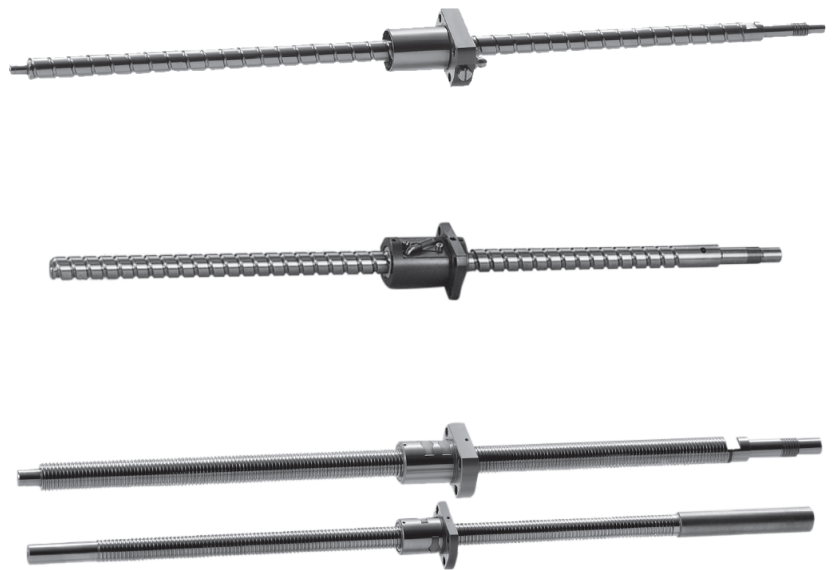
Parameters	Min	Max
Rail Width	8mm	12mm
Assembly Height	11mm	20mm
Dynamic Load	985 N	4500 N
Static Load	2630 N	11300 N

3.3 Ball Screw

NSK Ball Screws comes with minimum backlash without the application of preloading methods. The Gothic arc profile minimizes the clearance between ball and the grooves leading to backlash minimization.

Industries it caters to

Automobile Industries Manufacturer
Machine Tools Manufacturer
Automation Industries Manufacturer
Tyre Industry



3.3 Ball Screw (continued)

3.3.1 FINISHED SHAFT END COMPACT FA SERIES



Selection parameters

Parameters	Min	Max
Screw Diameter	6mm	25mm
Lead	8mm	50mm
Dynamic Load	550 N	12800 N
Static Load	715 N	32200 N

3.3.2 FINISHED SHAFT END VFA TYPE FOR TRANSFER EQUIPMENT



Selection parameters

Parameters	Min	Max
Screw Diameter	12mm	15mm
Lead	10mm	20mm
Dynamic Load	3750 N	7070 N
Static Load	6480 N	12800 N

3.3.3 FINISHED SHAFT END RMA TYPE AND BLANK SHAFT END RMS TYPE FOR TRANSFER EQUIPMENT



Selection parameters

Parameters	Min	Max
Screw Diameter	6mm	12mm
Lead	1mm	10mm
Dynamic Load	520 N	1350 N
Static Load	925 N	3190 N

3.4 Roller Pack

NSK Roller pack is a Linearly Rolling Guide in which the rollers can infinitely recirculate. It is used in large machine tools and some conveyor systems for heavy loads.

Selection parameters

Parameters	Min	Max
Width	25mm	45mm
Height	19mm	40mm
Length	65.5 N	138mm
Dynamic Load	31000 N	151000 N
Static Load	40500 N	191000 N



3.5 NSK Special Grease

It is an environmentally friendly and widely used grease for high load application. It is mineral oil based grease containing lithium thickener and several additives. It is superb in load resistance as well as stability in oxidization. It not only maintains good lubrication over a long period of time, but also demonstrates superb capability in retaining water. Even containing a large amount of water, it does not lose grease when it is softened.

Industries it caters to

Printing Press Industries
Automation Industry
Packaging Industry"

Product application

Printing Machines
Injection Molding Machine
Blow Molding Machine
SMT Machine
Fuji Machine
Packaging Machine

3.5.1 GREASE AS2



It is an environmentally friendly and widely used grease for high load application. It is mineral oil based grease containing lithium thickener and several additives. It is superb in load resistance as well as stability in oxidization. It not only maintains good lubrication over a long period of time, but also demonstrates superb capability in retaining water. Even containing a large amount of water, it does not lose grease when it is softened.

Selection parameters

Parameters	Min	Max
Temperature Range	-10°C	110°C

Industries it caters to

It is a standard grease for general NSK Linear Guides, Ball Screws, and Nanocarriers
It is prevalently used in many applications because of its high base oil Viscosity, High Load resistance, and stability in oxidization
Others

3.5.2 GREASE LR3



It contains a special synthetic oil for high temperature and stability, and a carefully selected anti-oxidation agent. This grease dramatically increases lubrication life under high temperature conditions. It is used for high speed, medium load. Lubrication life exceeded 2 000 hours in the endurance test at 150°C. Its rust prevention capacity in severe conditions such as water and moist environments is further strengthened.

Selection parameters

Parameters	Min	Max
Temperature Range	-30°C	130°C

Industries it caters to

It is a standard grease for Ball Screws PSS type (shaft dia. 15 mm or over), FSS type and VFA type
It is ideal for operation with medium load, at high speed such as positioning in High Tact material handling equipment
Others

3.5.3 GREASE LG2



This grease was developed by NSK to be exclusively used for Linear Guides, ball screws and Monocarriers in clean room. Compared to the fluorine grease which are commonly used in clean room, LG2 has several advantages such as, higher in lubrication function, longer lubrication life, more stable torque (resistant to wear) and higher rust prevention. In dust generation, LG2 is more than equal to fluorine grease in keeping dust volume low. Since the base oil is not a special oil but a mineral oil, LG2 can be handled in the same manner as general greases.

Selection parameters

Parameters	Min	Max
Temperature Range	-20°C	70°C

Industries it caters to

It is a standard grease for general NSK Linear Guides, Ball Screws, and Nanocarriers
It is prevalently used in many applications because of its high base oil Viscosity, High Load resistance, and stability in oxidization
Others

3.5 NSK Special Grease (continued)

3.5.4 GREASE LGU



This is a proprietary urea grease of NSK featuring low dust emission exclusively for ball screws and Linear Guides which are used in clean rooms. In comparison with fluorine base grease, which has been used commonly in clean rooms, LGU has better lubricating property, longer duration of lubricant, better torque variation, much better anti-rust property, and equivalent

Selection parameters

Parameters	Min	Max
Temperature Range	Minus 30°C	120°C

Industries it caters to

This is exclusive lubrication grease for ball screws and Linear Guides that are installed in equipment that requires cleanliness, as same as LG2 grease, and it can be used in high temperature range. Wash the linear guide and ball screws to remove oil prior to applying Clean Grease LGU, so the grease functions are fully utilized.

Clean grease is exclusively used for clean environments under normal pressure.

Others

3.5.5 GREASE PS2



The major base oil component is synthetic oil with mineral oil. It is an excellent lubrication especially for low temperature operation. It is for high speed and light load.

Selection parameters

Parameters	Min	Max
Temperature Range	Minus 50°C	110°C

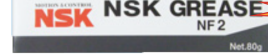
Industries it caters to

It is a standard grease for NSK miniature Linear Guides and ball screws.

It is especially superb for low temperature operation, but also functions well in normal temperatures, making it ideal for small equipment with light load.

Others

3.5.6 GREASE NF2



It uses high-grade synthetic oil as the base oil and urea base organic compound as the thickener.

It has remarkable anti-fretting corrosion property. It can be used in wide temperature range, from low to high, and has superior lubrication life.

Selection parameters

Parameters	Min	Max
Temperature Range	Minus 40°C	100°C

Industries it caters to

This grease is suitable for ball screws and Linear Guides of which application include oscillating operations.

Others

3.5 NSK Special Grease (continued)

3.5.7 GREASE NS7



It has remarkable torque characteristics (low torque), and can be used wide temperature range.

Selection parameters

Parameters	Min	Max
Temperature Range	Minus 40°C	130°C

3.5.8 GREASE NSL



It has anti-fretting property and superb torque characteristics (low torque) for high-speed operation.

Selection parameters

Parameters	Min	Max
Temperature Range	Minus 40°C	130°C

3.6 Hand Grease Pump

NSK hand Grease Pump is one of the best pump to lubricate all types of critical assemblies of LM Guideways, Ball Screw, and all other types of machine assemblies. Other features of this pump are

- Lite weight
- leakproof
- And it comes with 5 different nozzles attachments.

Selection parameters

Range Parameters	Value
Body Diameter	38.1 mm
Mass	240 g



Industries it caters to

Machining Centers
NC Lathes
Heavy Cutting Machine Tools
Packaging/Packing Machines
Others

Product application

Grease Hand Pump is used to apply grease in grease fittings for Linear Guides and ball screws etc
Others

3.7 Nozzles

The Nozzles are attached to Hand Grease Pump for the application of grease. Different nozzles are used for different types of grease fittings, also used for Linear Guides with no grease fittings.

Industries it caters to

Machining centers
NC Lathes
Heavy Cutting Machine Tools
Packaging/Packing Machines
Others

Product application

Can be used with grease fitting A, B, and C under JIS B1575 standard
Others

3.7.1 NSK NOZZLE NZ 1



Selection parameters

Parameters	Value
Length	220mm

Product application

Can be used with grease fitting A, B, and C under JIS B1575 standard
Others

3.7.2 NSK NOZZLE NZ 2



Selection parameters

Parameters	Value
Length	135mm

Product application

Can be used with grease fitting A, B, and C under JIS B1575 standard
However, there is no need to press the hand pump because the grease fitting and the nozzle come to contact due to the chucking mechanism at the tip
Others

3.7.3 NSK NOZZLE NZ 3



Selection parameters

Parameters	Value
Length	155mm

Product application

Dedicated for the – $\phi 3$ Drive-in Grease Fitting.
Others

3.7.4 NSK NOZZLE NZ 4



Selection parameters

Parameters	Value
Length	136mm

Product application

Used for Linear Guides and ball screws which do not have grease fitting
Supplies grease directly to the ball grooves, or through the opening of ball slide or ball slide to inside
Others

3.7.5 NSK NOZZLE NZ 5



Selection parameters

Parameters	Value
Length	300mm

Product application

The tip of the flexible nozzle is chuck nozzle.
Used to supply grease to the area where hand cannot reach
Others

3.7.6 NSK NOZZLE NZ 6



Selection parameters

Parameters	Value
Length	300mm

Product application

Flexible extension pipe connects the grease pump and the nozzle
Others

3.7 Nozzles (continued)

3.7.7 NSK NOZZLE NZ 7



Selection parameters

Parameters	Value
Length	317mm

Product application

Straight extension pipe connects the grease pump and the nozzle
Others

3.7.8 NSK NOZZLE NZ 8



Selection parameters

Parameters	Value
Length	135mm

Product application

For MCH Series Grease Replenishment
Others

3.8 Special Environments (SPACEA™ Series)

NSK's SPACEA™ Series bearings for special environments have a wide array of product variation applicable to vacuum environments, corrosive environments, clean environments, sanitary environments, high-temperature environments, dust-contaminated environments and non-magnetic requirement. The SPACEA™ Series offers high quality and high performance in severe operating environments, throughout a wide range of applications and in all kinds of machines and apparatuses.

Industries it caters to

Food Industry
Pharmaceutical
Others

Product application

Food Processing Machinery
Pharmaceutical Manufacturing Equipment
Machine Tools
Automotive
Others



3.8 Special Environments (SPACEA™ Series) (continued)

3.8.1 FOOD GRADE GREASE-PACKED BEARING



NSK SPACEA bearings are developed for special environments like food grade suitable environment, sanitary environment, high humidity environment etc. The application areas are Food industry, Pharmaceutical industry, Chemical industry, Steel industry etc.

Selection parameters

Parameters	Min	Max
Bore Diameter	4mm	40mm
Outside Diameter	9mm	80 mm
Limiting Load	21 N	1240 N

3.8.2 STAINLESS STEEL BEARINGS



Stainless steel bearings, the standard products of the NSK SPACEA™ Series for special environments, are suitable for high-humidity environments.

Selection parameters

Parameters	Min	Max
Bore Diameter	1mm	60mm
Outside Diameter	3mm	110mm
Limiting Load	4 N	2220 N
Dynamic Load Rating	81 N	44500 N

3.8.3 STAINLESS STEEL ANGULAR CONTACT BALL BEARINGS



- Outperforms standard bearing steel in terms of corrosion resistance.
- Achieves high running accuracy to ISO tolerance class P5.
- Supplied as bearings for universal matching with light preload when mounted in a face-to-face (DF) arrangement or back-to-back (DB) arrangement.
- Stainless steel angular contact ball bearings suitable for cleanroom environments in normal atmospheric conditions. Stainless steel angular contact ball bearings for cleanroom, vacuum, and high-temperature environments. Suitable for use in vacuum equipment or cleanroom applications operating under high-temperature conditions.

Selection parameters

Parameters	Min	Max
Bore Diameter	6mm	30mm
Outside Diameter	17mm	47mm
Limiting Load	86 N	650 N
Dynamic Load Rating	1730 N	13100 N

3.8 Special Environments (SPACEA™ Series) (continued)

3.8.4 STAINLESS STEEL SELF ALIGNING BALL BEARINGS



- Highly resistant to corrosion through the use of ES1 highly corrosion-resistant stainless steel.
- Self-aligning with the ability to accommodate misalignment of the axis and housing ranging from 4 to 7 degrees.

Selection parameters

Parameters	Min	Max
Bore Diameter	10mm	25mm
Outside Diameter	30mm	52mm
Limiting Load	230 N	520 N
Dynamic Load Rating	4750 N	10400 N

3.8.5 MOLDED-OIL BEARINGS



Molded-Oil Bearings are lubricated with NSK's original oil-impregnated material, Molded-Oil, and are suitable for corrosive and dust-contaminated environments.

Selection parameters

Parameters	Min	Max
Bore Diameter	10mm	30mm
Outside Diameter	22mm	55mm
Applied Load	23 N	590 N

3.8.6 CORROSION-RESISTANT COATED BEARINGS



Corrosion-resistant coated bearings (Nickel coating) are coated with a nickel coating on the outer and inner rings to enhance corrosion resistance and durability, and are suitable for corrosive environments such as normal atmosphere or high temperature.

Selection parameters

Parameters	Min	Max
Bore Diameter	10mm	30mm
Outside Diameter	26mm	55mm
Limiting Load	78 N	230 N

3.8.7 ESZ BEARINGS



ESZ Bearings have corrosion-resistant, and are superior to martensitic stainless steel. The bearings are having corrosion resistance equal to precipitation hardening type stainless steel and the hardness is improved more than 30%. The bearings are suitable for corrosive environment operating under atmospheric conditions.

Selection parameters

Parameters	Min	Max
Bore Diameter	10mm	60mm
Outside Diameter	26mm	110mm
Limiting Load	78 N	890 N

3.8.8 ESA BEARINGS

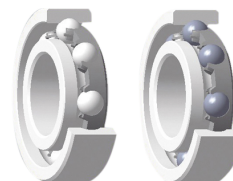


ESA bearings, combining austenite stainless steel and hardened surface layers, possess high hardness, corrosion resistance and non-magnetic properties, and are suitable for corrosive environments and non-magnetic requirement in normal atmosphere and vacuum.

Selection parameters

Parameters	Min	Max
Bore Diameter	8mm	30mm
Outside Diameter	22mm	55mm
Limiting Load	56 N	230 N

3.8.9 ALL-CERAMIC BEARINGS



With ceramic outer/inner rings and balls, all-ceramic bearings have self-lubricating fluororesin cages and are suitable for corrosive environments and non-magnetic requirement from normal atmosphere up to vacuum.

Selection parameters

Parameters	Min	Max
Bore Diameter	8mm	40mm
Outside Diameter	22mm	68mm
Limiting Load	73 N	820 N

3.8 Special Environments (SPACEA™ Series) (continued)

3.8.10 AQUA-BEARINGS™



Aqua-bearing features a special fluororesin for outer/inner rings and cages equipped to meet a broad range of applications in water, alkali and strong acid environments. Aqua-bearing is suitable for corrosive environments exclusively in normal atmosphere.

Selection parameters

Parameters	Min	Max
Bore Diameter	10mm	25mm
Outside Diameter	22mm	47mm
Limiting Load	22 N	100 N

3.8.11 DL2 GREASE PACKED BEARINGS



DL2 clean grease-packed stainless steel bearings are suitable for clean environments from normal atmosphere up to vacuum.

Selection parameters

Parameters	Min	Max
Bore Diameter	3mm	40mm
Outside Diameter	6mm	80mm
Applied Load	8 N	1240 N

3.8.12 DFO BEARINGS



Newly developed specification DFO bearings that take advantage of clean lubrication coatings: V-DFO and E-DFO. The V-DFO specification uses low vapor pressure fluorinated lubricant while the E-DFO specification uses a low vapor pressure hydrocarbon lubricant. Both specifications are applied to the inner and outer rings, balls, and cage to deliver superior cleanliness and long service life. The bearings are suitable for cleanroom environments ranging from normal atmospheric conditions to vacuum conditions.

Selection parameters

Parameters	Min	Max
Bore Diameter	4mm	40mm
Outside Diameter	9mm	80mm
Limiting Load	10 N	490 N

3.8.13 YS BEARINGS WITH SPACER



YS bearings for clean environments have newly developed self-lubricating cage, delivering high cleanliness and long life. These bearings are suitable for clean environments from normal atmosphere up to vacuum.

Selection parameters

Parameters	Min	Max
Bore Diameter	4mm	45mm
Outside Diameter	9mm	80mm
Limiting Load	10 N	490 N

3.8.14 KPM GREASE PACKED BEARINGS



These high-temperature bearings are grease-packed with NSK's long-life, high-temperature grease KPM, for use in normal atmosphere only.

Selection parameters

Parameters	Min	Max
Bore Diameter	4mm	40mm
Outside Diameter	9mm	80mm
Limiting Load	27 N	1240 N

3.8.15 SJ BEARINGS



SJ bearings have a "peapod" structure, with solid lubricant spacer joints mounted between two balls in cage pockets. These bearings are suitable for high-temperature environments from normal atmosphere up to vacuum.

Selection parameters

Parameters	Min	Max
Bore Diameter	8mm	30mm
Outside Diameter	22mm	55mm
Limiting Load	56 N	220 N

3.9 Super Precision Bearings

NSK has established groundbreaking techniques for the evaluation of inclusions in steel and for steel-making. We have developed materials that extend the life and enhance the performance of rolling bearings. Two types of materials are used for NSK Super Precision Bearings SHX Steel that endures ultra highspeed rotation, and Z Steel that achieves an extended operation life under normal conditions for a wide range of applications.

3.9.1 ANGULAR CONTACT BALL BEARINGS



NSK Super precision bearings are further divided as Angular Contact Ball Bearings, Cylindrical Roller Bearings, Angular Contact Thrust Ball Bearings, Ball Screw Support Bearings, Tapered Roller Bearings, Thrust Bearings and Angular Contact Ball Bearings ROBUSTSLIM.

3.9.2 CYLINDRICAL ROLLER BEARINGS



NSK Cylindrical roller bearings (CRB) are the feature higher radial rigidity and can take higher radial loads than ball bearings; this makes them perfectly suitable for applications that require high rigidity such as lathe spindles, Milling machine spindle, and for use as rear-end bearings subjected to large belt loads. The optimized internal design and cage shape of NSK cylindrical roller bearings are the reason why they achieve low heat generation and stable operation in high-speed applications. Double-row cylindrical roller bearings are also available in the "E44" specification for oil lubrication with lubrication holes and oil the groove in the center of the outer ring.

3.9.3 ANGULAR CONTACT THRUST BALL BEARINGS



NSK Angular Contact Thrust Ball Bearings are the Superior high-speed capability and high rigidity bearings. This fulfills the requirements of sustaining axial loads in machine tool spindles. For such types of requirements, NSK offers three types of bearings for customers. And they can choose it based on the structure and the characteristics of there machine. All of these bearings are designed to use in the combination of cylindrical roller bearings. These bearings are manufactured with special outer ring diameter tolerances to provide clearance between the outer diameter of the bearing and the inner diameter of the housing when mounted so that any radial load is supported by the cylindrical roller bearings only. The high-speed angular contact thrust ball bearings of the NSK ROBUST series are capable of high-speed performance while maintaining high rigidity. Contact angles of 40° (BTR) and 30° (BAR) result in superior high-speed performance and low heat generation.

Selection parameters

Parameters	Min	Max
Bore Diameter	5mm	420mm
Outside Diameter	16mm	620mm
Basic Dynamic Load	1.35 KN	675 KN
Basic Static Load	0.445 KN	1260 KN
Permissible Axial Load	0.345 KN	780 KN

Selection parameters

Parameters	Min	Max
Bore Diameter	25mm	800mm
Outside Diameter	47mm	1060mm
Basic Dynamic Load	19.7 KN	5700 KN
Basic Static Load	19.6 KN	16500 KN

Selection parameters

Parameters	Min	Max
Bore Diameter	50mm	380mm
Outside Diameter	80mm	560mm
Basic Dynamic Load	14.7 KN	425 KN
Basic Static Load	27.7 KN	1600 KN
Permissible Axial Load	12.6 KN	965 KN

3.9 Super Precision Bearings (continued)

3.9.4 BALL SCREW SUPPORT BEARINGS



NSKHPS:Reliability has been improved by focusing on material cleanliness, which has the biggest impact on bearing life, by employing NSK's proprietary material evaluation technology. The dynamic load rating has been improved by 5% compared with that of conventional bearings. The NSKTAC C Series features high axial rigidity and is suitable for machine tool feeding mechanisms, while the NSKTAC 03 Series with its high axial load capacity is well suited for the support of large ball screws in high-load drive applications such as electric injection molding machines. With these series users can achieve much lower torque and higher accuracy than with roller bearings.

3.9.5 TAPERED ROLLER BEARINGS



Tapered roller bearings are designed so that the apices of the cones formed by the raceways of the cone and cup and the conical rollers all coincide at one point on the central axis of the bearing. When a radial load is imposed, an axial force also occurs; therefore, it is necessary to use two bearings in opposition or some other multiple arrangement. The suffix J after the basic number of some high load capacity bearings (HR series) indicates that the cup back face raceway diameter, cup width, and contact angle conform to ISO specifications. Therefore, the cone assembly and cup of bearings with the same basic number suffixed with J are internationally interchangeable.

3.9.6 THRUST BEARINGS



Thrust ball bearings can sustain axial loads but no radial loads. For single-direction thrust ball bearings, pressed steel cages and machined brass cages are usually used. For identical bearing numbers, different cage types may imply a different number of balls; in such a case, the load rating will differ from the one listed in the bearing tables.

Selection parameters

Parameters	Min	Max
Bore Diameter	12mm	180mm
Outside Diameter	35mm	380mm
Basic Dynamic Load	18.5 KN	2520 KN
Basic Static Load	17.6 KN	6800 KN

Selection parameters

Parameters	Min	Max
Bore Diameter	100mm	360mm
Outside Diameter	150mm	480mm
Basic Dynamic Load	117 KN	1510 KN
Basic Static Load	205 KN	2910 KN

Selection parameters

Parameters	Min	Max
Bore Diameter	300mm	2530mm
Outside Diameter	47mm	2750mm
Basic Dynamic Load	20.6 KN	5750 KN
Basic Static Load	42 KN	60500 KN

3.9.7 ANGULAR CONTACT BALL BEARINGS ROBUSTSLIM



Selection parameters

Parameters	Min	Max
Bore Diameter	130mm	400mm
Outside Diameter	165mm	500mm
Basic Dynamic Load	20.2 KN	139 KN
Basic Static Load	91 KN	830 KN
Permissible Axial Load	28.5 KN	221.6 KN

A series of angular contact ball bearings that has been designed specifically for swiveling spindle heads used in machine tools, and is suitable for high-precision processing with multi-axis machines. The ROBUSTSLIM series of high accuracy low-profile angular contact ball bearings offer high rigidity and low torque.

3.10 Grease Rolling Bearings

The NSK Grease lineup is characterized by excellent water-resistance, excellent corrosion-resistance, low-evaporation, and a wide functional temperature range (-10 to 110). The lineup, which features lithium-soap thickened, mineral oil-based grease, includes two multi-purpose greases designed for use under normal operating conditions and one grease designed for heavy-load scenarios

3.10.1 NSKGM2



NSK has developed special grease for rolling bearings for MRO requirements. The NSKGM2 grade is used for multi-purpose use in industrial machinery.

Selection parameters

Parameters	Min	Max
Temperature Range	Minus 10°C	110°C

3.10.2 NSKGM3



NSK has developed special grease for rolling bearings for MRO requirements. The NSKGM3 grade is used for multi-purpose use in industrial machinery.

Selection parameters

Parameters	Min	Max
Temperature Range	Minus 10°C	110°C

3.10.3 NSKGM3



NSK has developed special grease for rolling bearings for MRO requirements. The NSKGP2 grade is used for heavy load applications in industrial machinery.

Selection parameters

Parameters	Min	Max
Temperature Range	Minus 10°C	110°C

3.11 Roller Bearings

NSK Roller bearings consists Cylindrical roller bearing, Spherical roller bearing, Tapered roller bearing, Thrust bearing and Needle roller bearing for wide range of applications.

Industries it caters to

Steel Industry
Mining & Construction
Machine Tools
Automotive
Others

Product application

Motor Vehicles
Construction and Cargo Handling Equipment
Agriculture Machinery
Business Machines
Hydraulic Machines
Others



3.11 Roller Bearings (continued)

3.11.1 CYLINDRICAL ROLLER BEARING



NSK Roller bearings are further divided into Cylindrical roller bearing, Spherical roller bearing, Tapered roller bearing, Thrust bearing and needle roller bearing. These are used in different industries like steel industry, Mining and construction industry, automotive industry etc.

3.11.2 SPHERICAL ROLLER BEARING



Types EA, C, CD, CA, which are designed for high load capacity, are available. Types EA, C, and CD have pressed steel cages; and type CA has machined brass cages. Type EA has especially high load capacity, features low torque, and has a high-strength cage. An oil groove and holes are provided in the outer ring to supply lubricant and the bearing numbers are suffixed with E4. To use bearings with oil grooves and holes, it is recommended to provide an oil groove in the housing bore, since the depth of the groove in the bearing is limited.

3.11.3 TAPERED ROLLER BEARING



Bearings of this type use conical rollers guided by a back-face rib on the cone. These bearings are capable of taking high radial loads and axial loads in one direction. In the HR series, the rollers are increased in both size and number giving it an even higher load capacity. They are generally mounted in pairs in a manner similar to single-row angular contact ball bearings. In this case, the proper internal clearance can be obtained by adjusting the axial distance between the cones or cups of the two opposed bearings. Since they are separable, the cone assemblies and cups can be mounted independently.

Selection parameters

Parameters	Min	Max
Bore Diameter	20mm	500mm
Basic Load Ratings Cr	14300 N	298000 N
Basic Load Ratings CDr	12700 N	585000 N

Selection parameters

Parameters	Min	Max
Bore Diameter	20mm	1400mm
Basic Load Ratings Cr	29300 N	24700 KN
Basic Load Ratings CDr	26900 N	65000 KN

Selection parameters

Parameters	Min	Max
Bore Diameter	12mm	440mm
Basic Load Ratings Cr	10300 N	7100 KN
Basic Load Ratings CDr	8900 N	4200 KN

3.11.4 THRUST BEARING



Available with cylindrical or spherical rollers, NSK's thrust roller bearings sustain only axial loads, but they are suitable for heavy loads and have high axial rigidity. Spherical thrust roller bearings contain convex rollers and have a self-aligning capability and are free of any influence of mounting error or shaft deflection. Standard cages are machined brass, where the E-type offers a pressed cage for high load capacity.

Selection parameters

Parameters	Min	Max
Bore Diameter	10mm	500mm
Basic Load Ratings Cr	10100 N	8000 KN
Basic Load Ratings CDr	14000 N	33000 KN

3.11.5 NEEDLE ROLLER BEARING



Needle roller bearings are a bearing that is classified as either radial or thrust depending on the direction of the load being supported. Needle roller bearings include bearings whose rollers slightly exceed the size range of needle rollers as stipulated by ISO. Needle roller bearings include drawn cup and solid radial bearings, as well as application-specific cam followers, and roller followers. Thrust bearings include thrust needle bearings.

Selection parameters

Parameters	Min	Max
Bore Diameter	3mm	490mm
Basic Load Ratings Cr	890 N	1980 KN
Basic Load Ratings CDr	1220 N	4750 KN